

Nara Ávila

Master's Student

Department of Mathematical Physics
Physics Institute, University of São Paulo (USP)

E-mail: nara.moraes@usp.br

Website: <https://fma.if.usp.br/~avilamrs>

Education

M.Sc. in Nuclear Physics [\[link\]](#)

2025 – Present

Physics Institute (IF)

University of São Paulo (USP)

B.Sc. in Computer Engineering [\[link\]](#)

2006 – 2011

São Carlos School of Engineering (EESC)

University of São Paulo (USP)

The interdisciplinary curriculum encompassed four major areas of computer engineering:

Computer Science & Software Engineering: Training in the theoretical and practical foundations of computing, spanning algorithms and data structures, theory of computation and compilers, as well as object-oriented programming, software engineering, databases, distributed systems, performance evaluation, and computer graphics.

Computer Architecture: Study of computer systems from digital organization to processor-level architecture, including microprocessors, digital systems design, and hardware–software interfacing.

Physics & Electronics: Foundation in classical and quantum mechanics, electromagnetism, and experimental physics, integrated with the study of electric and electronic circuits, semiconductor physics, microelectronics, microcontrollers, signal processing, control systems, and analog and digital integrated circuit design.

Communications & Networks: Study of information transmission and networked systems, covering communication systems and digital communications, antennas and propagation, optical communications, computer networks, high-performance networking, and mobile network architectures.

Academic Positions

Research Collaborator [\[link\]](#)

2023 – 2025

40 hours/week

Project: *Quantum Computing Applied to Molecular Modeling*

Department of Materials Physics and Mechanics (DFMT)

Institute of Physics (IF), University of São Paulo (USP)

Research Context: Investigation of quantum computing methods for first-principles molecular modeling in materials and energy research. The work focused on computing ground-state energies of molecules relevant to the energy transition, including H₂ and CO₂, as a first step toward the use of quantum simulations in the discovery and development of new materials and sustainable energy technologies.

Quantum Simulation of Molecular Systems: Developed first-principles quantum simulations for molecular ground-state energy estimation. Electronic-structure problems were formulated within the Born–Oppenheimer approximation and second quantization, with fermionic Hamiltonians mapped onto qubit operators using the Jordan–Wigner transformation. Ground-state energies were estimated with VQE using UCCSD ansätze and benchmarked against Hartree–Fock and Full Configuration Interaction (FCI) calculations. The study examined basis-set selection, active-space approximations, shot counts, and qubit-reduction strategies, and explored restricted Boltzmann machines (RBMs) as alternative variational representations of molecular quantum states.

Quantum Computing Methods & Platforms: Implemented and evaluated gate-based quantum algorithms across Amazon Braket, IBM Qiskit, Google Cirq, CERN Qibo, and PennyLane, considering superconducting, trapped-ion, photonic, and neutral-atom architectures. Hardware-oriented studies examined execution time, noise, and decoherence, together with NISQ error mitigation using zero-noise extrapolation (ZNE) and Clifford data regression (CDR).

Teaching Assistant [\[link\]](#)

Fall 2009

8 hours/week

- SME0300 Numerical Methods I

Institute of Mathematics and Computer Sciences (ICMC)

University of São Paulo (USP)

Teaching Assistant [\[link\]](#)

Spring 2009

8 hours/week

- SME0601 Numerical Methods II

Institute of Mathematics and Computer Sciences (ICMC)

University of São Paulo (USP)

Academic Awards

- Best Presentation in Environmental Management System (SHS0436) 2009
 - Recognized for an impactful final course presentation entitled “Harmonic and Disharmonic Systems,” due to its combination of visual simplicity and strong conceptual content. Using images alone, the presentation explored in detail the diversity, complexity, and beauty of harmonic systems in nature—including coral ecosystems and harmonious ecological relationships such as mutualism, commensalism, and epiphytism—in contrast with human systems that disrupt environmental balance, illustrated through examples of waste accumulation, pollution, environmental degradation, and pollution-related wildlife mortality, alongside other disharmonic relationships also found in nature, such as parasitism and disease. Received the maximum final grade (10.0/10.0).
- 9th place in the ACM-ICPC / SBC Brazilian Programming Contest [\[link\]](#) Aug 2008
 - Competed as a member of team “Greve de Acertos” in the São Paulo State round of the Brazilian programming competition organized by the Brazilian Computer Society (SBC) as part of the International Collegiate Programming Contest (ICPC).
- 6th place in the IPhO / SBF Brazilian Physics Olympiad [\[link\]](#) 6th – 2003
 - National physics olympiad organized by the Brazilian Physical Society (SBF) and part of Brazil’s selection process for the International Physics Olympiad (IPhO).
- 18th place in the OBM / XI OMEG [\[link\]](#) 18th – 2002
 - Mathematics olympiad competition associated with the Brazilian Mathematical Olympiad (OBM), Brazil’s national mathematics olympiad and part of the country’s selection pathway for the International Mathematical Olympiad (IMO).

Grants**Entrepreneurship Workshop (PRPG/USP)** [\[link\]](#)

2025

Project Title: StrategiQ — AI-Based SaaS for Quantum Computing Governance Consulting
Role: Founder and Project Lead of a three-member team conducting market research and developing the business plan for StrategiQ developed through the Scientist Entrepreneur Training (DPG5011)

Funding Agency: Office of the Provost for Graduate Studies, University of São Paulo (PRPG/USP)

Total Amount: R\$3,720.00

132734/2025-7 [\[link\]](#)

2025–2027

Program: Master’s Scholarship (GM)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Host Institution: University of São Paulo (USP)

Total Amount: R\$50,400.00

409642/2022-3 [\[link\]](#)

2023–2024

Project Title: Quantum Computing Applied to Molecular Modeling

Program: Technological Development in ICTs (DTC-F)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Scholarship Process: 301681/2023-6

Total Amount: R\$36,300.00

Software

1. MORAES, N. A.; LUZUM, M. W. *hic-bayes*. [\[link\]](#)

2026

Python software for Bayesian parameter estimation and Gaussian-process emulation in relativistic heavy-ion collision studies. Implements PCA-based dimensionality reduction, Gaussian-process emulators, multivariate likelihoods, MCMC sampling, posterior analysis, and multi-fidelity workflows.

2. MORAES, N. A. *Quantum Battleship*. [\[link\]](#) 2025

Quantum computing game that allows users to play Battleship using real quantum computers.

3. MORAES, N. A. *Zotero Mirroring Plugin*. [\[link\]](#) 2026

Zotero plugin for mirroring the hierarchical structure of a Zotero library onto the local filesystem.

4. MORAES, N. A. *GitHub Release and Milestone CLI*. [\[link\]](#) 2026

Command-line tool for automating GitHub releases, milestones, and software development cycles.

5. MORAES, N. A. *Molecular Symmetry*. [\[link\]](#) 2025

Web application for parsing molecular geometries from XYZ files and generating molecular symmetry reports.

Leadership Experience

Founder & Team Lead 2025

DPG5011 – Entrepreneurship, University of São Paulo (USP)

- Founded and led a six-member interdisciplinary team in the development of an innovative startup focused on quantum computing.
- Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.

Undergraduate Research Mentor 2025

Institute of Physics, University of São Paulo (IFUSP)

- Mentored a group of five undergraduate students in scientific initiation projects in quantum computing, providing guidance throughout their research activities.
- Supervised the development of individual research projects presented by the students at the Undergraduate Research Symposium.

Professional Diplomas

Postgraduate Specialization in Financial and Capital Markets [\[link\]](#) 2021 – 2022

- Quantitative methods, corporate finance, financing, and financial products and markets
- Mackenzie Presbyterian University*

Professional Experience

Software Engineering Specialist 2020 – 2022

Open Co

- Developed and optimized backend services for a digital banking platform, including the first installment payment solution via PIX.
- Proposed and prototyped the migration of a credit-analysis workload from Kubernetes (EKS) to AWS Lambda, reducing response time and operational costs by approximately 90%.

Senior Software Engineer 2019 – 2021

TIDEXA

- Contributed to the migration of a payroll system from Oracle Forms/PL-SQL to a Java-based platform.
- Led technical decisions and development activities within the engineering team.

Competitive Public-Sector Examination Preparation 2016 – 2019

Independent, full-time study

- Multidisciplinary preparation for competitive Brazilian public-sector examinations, including Law, Accounting, Economics, Public Finance, Public Administration, and Business Administration.

Mid-Level Software Engineer 2014 – 2016

Itautec – Grupo Itautec

- Redesigned a business-critical software application using software design patterns and engineering best practices, improving usability, reliability, and maintainability. The application was deployed across more than 5,000 branches of a major Brazilian bank.

Junior Software Engineer

2012 – 2014

Itautec – Grupo Itautec

- Developed and maintained software applications for retail industry.

Talks & Presentations

Conference Presentations

1. “The Effect of Correlated Experimental Uncertainties on Bayesian Inference of Quark–Gluon Plasma Properties.” Poster presentation at the PHENOEXP Workshop, August 2026. [\[link\]](#).
2. “Study of the Effect of Correlated Experimental Uncertainty on Bayesian Inference of Quark–Gluon Plasma Properties.” Oral presentation at the XXXIII Reunião de Trabalho sobre Interações Hadrônicas (RETINHA XXXIII), June 2026. [\[link\]](#).
3. “Risk Management and IT Governance in the Adoption of Quantum Computing: A COBIT 5-Based Conceptual Model.” Oral presentation at the 5th Congress on Risk Management and ERM, FEA–USP, December 2025. [\[link\]](#).
4. “Exploring the Potential of Quantum Computing in Materials Science.” Oral presentation (co-author), Autumn Meeting of the Brazilian Physical Society, May 2025. [\[link\]](#).

Industry & Professional Talks

1. “Perspectives on Quantum Computing.” Talk at Rio Innovation Week – AI and Meta World, 2024. [\[link\]](#).
2. “Quantum Computing. Hands On Quantum Search Algorithms.” Talk at Open Co Tech Day, 2022.
3. “Shift Your Programming Paradigm to a Quantum Mindset.” Talk at The Developer’s Conference (TDC Business), 2022. [\[link\]](#).
4. “Fundamentals of Quantum Computing.” Talk at Open Co Tech Day, 2021.
5. “Technical Software Documentation.” Seminar at ITAUTEC, 2014.

Selected Highlights

School Environmental Research Project – Municipal Waste Management

1999

Goiânia, Goiás, Brazil

- Following a field visit to the Goiânia Municipal Landfill, interviewed Pedro Wilson Guimarães, then Federal Deputy for the State of Goiás, for a school research project on municipal solid waste management.
- Discussed waste-management challenges and technologies, including landfill biogas recovery and international approaches to waste incineration.
- Wilson later served as Mayor of Goiânia (2001–2004), where his administration established municipal selective waste collection through legislation enacted in 2002.

Extracurricular Activities

1st Place – Brazil Interstate Taekwon-Do ITF Competition [\[link\]](#)

August 2023

- Two Gold medals in Sparring and Patterns (Tul).
- Buzios. Rio de Janeiro, Brazil.

1st Place – Children’s Drawing Contest [\[Art Portifolio\]](#)

October 2000

Claro – Telecommunications Company, Brazil

- First-place award in a Children’s Day drawing competition organized by Claro for employees’ children.